

Maciej Flaga

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📍 Rząska, Poland



PROFILE

Undergraduate student in Nanoengineering of Materials with a strong interest in computational physics, numerical simulations, and theoretical aspects of materials science. Experienced in C/C++, Python and CUDA for scientific computing. Seeking to develop skills in simulation, modeling, and data analysis.

EDUCATION



Student

📍 Kraków, Poland

Faculty of Physics and Applied Computer Science

Field: Nanoengineering of Materials

Specialization: Modeling

AGH University of Krakow

📅 Oct. 2023 – ongoing

SOFTWARE AND LANGUAGE SKILLS

Software



Proficient in C/C++



Proficient in Python (NumPy, Matplotlib, pandas)



Proficient in $\text{\LaTeX}$



Experienced with CUDA C++



Fairly experienced with Matlab



Basic knowledge of the Linux OS

Languages



Polish – Native



English – Fluent



German – Basic

Certificates



ACERT Business English C1 (2025)

ACADEMIC PROJECTS

Mass Transport Simulation

Developed a GPU-accelerated Computational Fluid Dynamics (CFD) engine to simulate incompressible flow and mass transport using the **Navier-Stokes** and **Advection-Diffusion** equations. Implemented the Stream Function-Vorticity formulation using Finite Difference Methods, solving the resulting linear systems via **parallelized Red-Black Gauss-Seidel relaxation** custom **CUDA** kernels. The simulation features a **Crank-Nicolson scheme** for stable time-stepping of mass distribution and integrates a **Python/FFmpeg** pipeline to render high-framerate visualizations of velocity fields and concentration heatmaps.

(2+1)D Wave Equation Simulation

Developed a high-performance visualizer for the (2+1)D wave equation by deriving an exact analytical solution via **separation of variables** and **Fourier series expansion**. Implemented the numerical evaluation of the double infinite series using **CUDA C++**, optimizing performance by parallelizing the summation loops on the GPU to handle high-resolution spatial grids. The project features a fully automated pipeline that orchestrates the C++ backend, **Python (Matplotlib)** for 3D surface plotting, and **FFmpeg** for video rendering using a custom **Makefile**.

JOB EXPERIENCE

Production line worker

📍 Modlniczka, Poland

YK Poland Sp. z o.o.

📅 Jul. 2024 – Sep. 2024 (3 months)

Executed the complete production cycle of metal photo frames, including profile cutting, assembly, shrink-wrapping, and final packaging.

Production line worker

📍 Augsburg, Germany

BMK Professional Electronics GmbH

📅 Jul. 2023 – Sep. 2023 (3 months)

Performed manual PCB assembly, operated the reflow soldering process, and handled board preparation involving masking and machine-assisted dispensing.

INTERESTS

Scientific

- Scientific computing

Non-scientific

- Climbing